

Article

Real-Time Household Waste Detection and Classification for Sustainable Recycling: A Deep Learning Approach

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Abstract: As global waste production continues to rise, improper handling of household waste significantly contributes to environmental pollution and resource depletion. Inefficient sorting at the household level leads to the contamination of recyclables, reducing recycling efficiency and increasing landfill waste. Effective waste sorting is essential for conserving manual labor, protecting the environment, and ensuring sustainable development for human progress. Recently, advancements in deep learning and computer vision have offered a promising pathway to improve the sorting process, though significant developmental steps are still required. Enhancing the efficiency of automated waste detection and classification through computer vision could bring substantial societal and environmental benefits. However, classifying and identifying waste materials presents challenges due to the complex and diverse nature of waste, coupled with the limited availability of data on waste management. This paper presents a real-time waste detection and classification system based on the YOLOv8 deep learning model, designed to enhance waste sorting processes at the household level. The proposed system detects and classifies a diverse range of household waste items. Experiments were conducted on a custom waste dataset comprising 3775 images across 17 types of common household waste. The one-stage YOLOv8 model demonstrated superior performance, outperforming traditional two-stage detectors. To improve the accuracy and robustness of the original YOLOv8, five data augmentation techniques and two attention mechanisms were incorporated. Notably, the enhanced YOLOv8-CBAM model achieved a mean average precision (mAP) of 89.5%, a significant improvement with a 4.2% increase over the baseline model. The methodology and improvements applied provide a more efficient and effective AI framework for real-time applications in smart bins, robotic waste pickers, and large-scale recycling systems.



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Keywords: waste detection; recycling; deep learning; CNN; YOLO; attention mechanisms

1. Introduction

In today's world, the growing population and industrialization have resulted in an unprecedented increase in waste production. It is projected that global waste production will rise by 20% by the year 2030, potentially reaching an annual output of approximately 2.6 billion tons [1]. By 2050, the world is expected to generate 3.40 billion tons of waste annually [1].

With rising living standards, the diversity of household waste types is also increasing [2]. Efficient waste management and recycling practices, especially at the household level, have become critical in mitigating the adverse effects of this waste on the environment [3]. By adopting sustainable waste disposal methods, households not only reduce the strain on landfills and prevent pollution but also contribute to reducing the demand for raw resources and energy consumption. Additionally, citizens' participation in the recycling

process is critical to ensuring the long-term sustainability of waste management systems. Numerous studies have identified a connection between a nation's Gross Domestic Product (GDP) and its recycling habits [4–7]. Although wealthier countries have made significant strides in encouraging citizens to actively participate in recycling efforts, they often fall short of meeting the ambitious goals set by environmental policies.

Despite the critical importance of proper waste separation for environmental sustainability, many individuals still neglect this practice, hindering efforts to achieve effective waste management. A major barrier to the successful implementation of regulations is the lack of understanding among residents regarding waste classification. Furthermore, low motivation to participate in waste sorting not only compounds the issue but also undermines the potential benefits of sustainable waste practices [8].

Waste sorting continues to be a major problem in society, characterized by diverse waste types, resource waste, and labor-intensive processes [9]. Traditional manual waste sorting methods were expensive, unreliable, and exposed workers to health risks from hazardous materials [10–12]. Furthermore, a labor shortage in the industry often led to the recruitment of untrained employees to manage the increasing volumes of urban waste [13]. These challenges underscore the urgent need for effective waste recycling strategies [14]. Consequently, city officials and governments are in critical need of automated, rapid, and straightforward waste classification and detection systems for use in garbage sorting centers [15,16].

Existing waste detection systems primarily rely on convolutional neural networks (CNNs) for object detection. Over the years, numerous CNN-based detection algorithms have been developed, offering varying degrees of accuracy and computational efficiency. For waste detection and classification, CNN algorithms are broadly categorized into two-stage and one-stage approaches [17]. Two-stage approaches, such as R-CNN and Faster-RCNN [18], have achieved significant success in traditional target detection tasks. However, they exhibit several shortcomings when applied to the specific domain of waste detection and classification. In a two-stage model, the process begins with the segmentation of candidate regions, followed by classification and localization. This multi-step procedure significantly increases the overall inference time, making the model less efficient for real-time applications [19]. Moreover, the diverse nature of waste items, coupled with the need to detect numerous small objects, increases computational complexity significantly. These limitations highlight the need for more efficient approaches to waste detection.

In contrast to two-stage methods, one-stage object detection algorithms, such as YOLO (You Only Look Once) and SSD (Single Shot MultiBox Detector), streamline the detection process by performing both object classification and localization in a single step. This eliminates the need for region proposals, making one-stage models significantly faster and more computationally efficient, especially suitable for real-time applications like waste detection [20,21]. By treating object detection as a regression problem, one-stage models predict class labels and bounding box coordinates simultaneously across predefined anchor boxes or grids, resulting in quicker processing times [20]. This speed advantage is critical in scenarios like household waste sorting systems, robotic waste pickers, and smart recycling bins [22]. Recent advancements, particularly in newer YOLO versions, have greatly improved the balance between speed and accuracy, making one-stage detectors increasingly competitive with two-stage models for complex tasks like waste classification and detection [23]. In addition, the customization and adaptability of YOLO models significantly strengthen their utility in complex environments such as waste management [24]. Tailoring YOLOv8 to recognize and classify varied waste types enhances sorting accuracy, facilitating more efficient waste processing operations. This adaptability

ensures robust performance across diverse conditions and material characteristics, thereby advancing recycling technologies and contributing to greater environmental sustainability.

This work proposes a deep learning framework for real-time household waste classification and detection to support sorting efficiency and accuracy in smart recycling applications. The framework is based on YOLOv8 and proposes the following main contributions.

1. A YOLOv8-based waste classification and detection model, trained on a custom 17-category dataset, overcomes previous dataset limitations and harnesses YOLOv8's computational efficiency for real-time use in robotic waste pickers and smart bins.
2. The model is enhanced with SE and CBAM attention mechanisms, which improve the detection of small waste items and reduce background noise across various scales and complexities.
3. A data balancing and augmentation strategy is implemented to enhance the overall detection performance, ensuring the model remains effective and deployable. The combination of the CBAM attention mechanism and data augmentation significantly improves the waste classification and detection accuracy in the YOLOv8 model.

The rest of this paper is organized as follows: Section 2 reviews relevant deep learning studies. Section 3 outlines the framework, detailing the composition and preprocessing of the image dataset, the augmentation strategies employed, and the configuration of the YOLOv8-based waste detection and classification system. Section 4 provides an intricate overview of the proposed integration of attention mechanisms into YOLOv8 to enhance its detection capabilities. Section 5 presents a comprehensive account of the experiments and conducts an in-depth analysis of the results obtained. Section 6 concludes the study. Section 7 suggests future research directions to address the complex challenges of industrial waste management.

2. Literature Review

Recently, the application of deep learning in waste management has attracted considerable attention [25–27]. In deep learning, CNNs have revolutionized computer vision tasks. Recent advancements in computing power and the development of complex CNN architectures have led to notable improvements in image recognition quality.

The progression of object detection methods has been shaped by two key approaches: two-stage detectors and single-stage detectors. Two-stage detectors, such as R-CNNs, have traditionally emphasized accuracy. These models break down the detection process into two separate stages: first, identifying regions of interest where objects might be present and then classifying these regions to detect objects. For example, R-CNN [28] initially proposes regions and then uses a classifier to confirm object presence, ensuring precise localization. Later models, like Fast R-CNN [29] and Faster R-CNN [30], streamlined this process by improving computational efficiency. They accomplished this by sharing convolutional layers and incorporating region proposal generation directly within the network. Despite the high accuracy achieved by two-stage detectors—especially in detecting smaller or partially obscured objects—these models are often computationally demanding and slower than their single-stage counterparts, making them less ideal for real-time applications [30].

Multiple studies have applied the traditional two-stage CNN detectors to detect and classify waste in different environments. For example, three different domestic waste classes were tested and documented in [31] using the Fast R-CNN model. For the overall classification of the waste images, the authors achieved a mean average precision (mAP) of 0.683. Li, Yan, and Xu (2020) applied the improved Mask Scoring R-CNN to classify waste into four categories: hazardous, kitchen, recyclable, and other garbage, achieving an accuracy of 65.8% [32]. Conley et al. (2022) proposed a deep learning-based approach

to monitor and quantify trash along urban roadsides using vehicle-mounted cameras [33]. Their trained Mask R-CNN outperformed the other models, achieving 77% accuracy. In the aquatic environment, the AquaVision system was developed by [34] to detect waste in water bodies, using one-stage RetinaNet and two-stage Faster-RCNN. Their proposed model classifies waste into four categories—glass, metal, paper, and plastic—achieving an mAP of 0.814. The one-stage RetinaNet outperforms Faster R-CNN in this application due to its ability to handle class imbalance more effectively and detect objects more efficiently in a single pass. In the construction field, Wang, Li, and Zhang (2019) introduce a construction waste recycling robot that uses Faster R-CNN to detect and collect three types of nails and screws with an accuracy of 0.891 [35]. The robot integrates computer vision and neural networks for path planning in unstructured environments.

In contrast, single-stage CNN detectors like YOLO [20] and SSD [21] were designed with speed in mind. These models bypass the region proposal step entirely, detecting objects in a single forward pass through the network. YOLO, for instance, divides the image into a grid and simultaneously predicts bounding boxes and class probabilities for each cell. This architecture allows YOLO and SSD to achieve real-time performance, which is critical for real-time waste detection. However, simplification comes at the cost of accuracy. Single-stage detectors typically struggle with complex scenes and small object detection [21], often exhibiting poorer performance when compared to two-stage detectors.

Nevertheless, significant improvements have been made in newer versions of YOLO, particularly in YOLOv8, which offers advancements in both speed and accuracy [23]. YOLOv8 integrates state-of-the-art architectural optimizations, such as decoupled head designs for improved performance on complex object detection tasks, and a more refined anchor-free mechanism that enhances the detection of smaller objects and complex backgrounds [36]. Compared to its predecessors, YOLOv8 features a more efficient backbone, which reduces computational cost while maintaining high detection precision. This version also leverages innovations in feature fusion, which contributes to improved object localization and classification accuracy [36]. As a result, YOLOv8 narrows the gap between single-stage and two-stage detectors by providing fast inference times without sacrificing much accuracy, positioning it as a superior choice for real-time applications that still require high precision.

Different studies have utilized the YOLO series for waste detection tasks, with most studies focusing on surface-level waste detection, while limited research has been conducted on municipal waste recycling. For example, Yang (2021) developed a YOLOv4-based model optimized for classifying waste in restaurant environments, addressing the challenges of complex backgrounds [37]. The model was trained to detect and classify five different recyclable item classes. It achieved a detection accuracy of 77.78%, outperforming other models such as SSD and Fast R-CNN in both detection efficiency and speed. Yang et al. (2022) developed an improved YOLOv5-based model for detecting floating garbage in rivers [38]. The model was trained on two datasets: a private dataset with 2400 images and five classes of floating garbage and the FloW-Img public dataset with 2000 images and five classes. Their trained model achieved an average precision of 90.85% on the private dataset and an average precision of 92.18% on the FloW-Img dataset. Building on previous research, Demetriou et al. (2023) developed a YOLOv7-based model for the real-time detection of construction and demolition waste (CDW) [39]. The model was trained on a dataset across three classes: brick, concrete, and tile. Comparative experiments were conducted between single-stage (YOLOv7) and two-stage (Faster R-CNN) detectors. YOLOv7 achieved a mean average precision (mAP_{50:95}) of approximately 70% and fast inference time, outperforming Faster R-CNN models in terms of speed and accuracy. Sujjada et al. (2023) developed a YOLOv8-based model for detecting and classifying seven types of plastics, including PET,

HDPE, PVC, LDPE, PP, PS, and others [40]. The model achieved an mAP_{0.5:0.95} of 93.38%. YOLOv8 outperformed other models in both speed and accuracy, proving highly efficient for real-time plastic detection across various scenarios.

While several studies have utilized two-stage and single-stage models for waste detection in specialized environments (e.g., roadside, aquatic, or construction sites), fewer have applied these models, particularly the newer versions like YOLOv8, to comprehensive municipal waste sorting, which involves a diverse range of household waste categories. Moreover, many existing studies using the YOLO series focus on specific waste types, such as plastic or construction materials, rather than addressing the broader variety of waste found in municipal settings. Despite advancements in real-time waste detection with models like YOLOv7 and YOLOv8, there remains a significant opportunity to explore their applications for detecting and classifying various household waste types to improve recycling efficiency on a larger scale.

As observed from the literature review, at the time of writing, there was limited emphasis on the application of the YOLOv8 model in waste detection, leaving open questions about its effectiveness on complex waste datasets and the possibilities for enhancing its architecture. This work aims to bridge this knowledge gap by investigating the advantages offered by the YOLOv8 model in classifying and detecting various waste types. Designed for real-time performance, the proposed model is ideally suited for practical applications in smart waste management systems and automated sorting processes.

3. Methodology

Creating accurate and efficient models to automatically detect and classify recyclable, non-recyclable, and hazardous materials is key to improving waste management systems [41]. Unlike image classifiers, which simply assign a class label to an entire image, object detectors present a greater challenge as they also require localizing objects within the image by drawing bounding boxes around them.

One effective solution for real-time detection and classification is using AI-powered pattern recognition technology, which has become highly valued for identifying and sorting waste materials [42]. This research focuses on using CNNs, particularly the recent YOLOv8 model developed by Ultralytics [36]. YOLOv8 brings notable improvements, such as better accuracy and speed for real-time detection. Its updated architecture, which includes the Feature Pyramid Network (FPN) and Path Aggregation Network (PAN), enhances multi-scale detection. Additionally, YOLOv8 offers faster and more efficient training and inference, making it highly adaptable for different tasks and applications. While YOLO models have been used in various fields, applying YOLOv8 to detect and classify 17 types of household waste is a leading solution for real-time waste management systems. The research methodology, shown in Figure 1, outlines a process where the collected images are annotated and preprocessed using Roboflow (<https://roboflow.com/>, accessed on 8 November 2024). Data augmentation is applied to improve diversity, after which the dataset is trained and tested using the YOLOv8 model in Google Colab. Model accuracy is evaluated, and if it meets the required threshold, the process ends. Otherwise, data improvement, model refinement and retraining are performed iteratively until the desired accuracy is achieved.

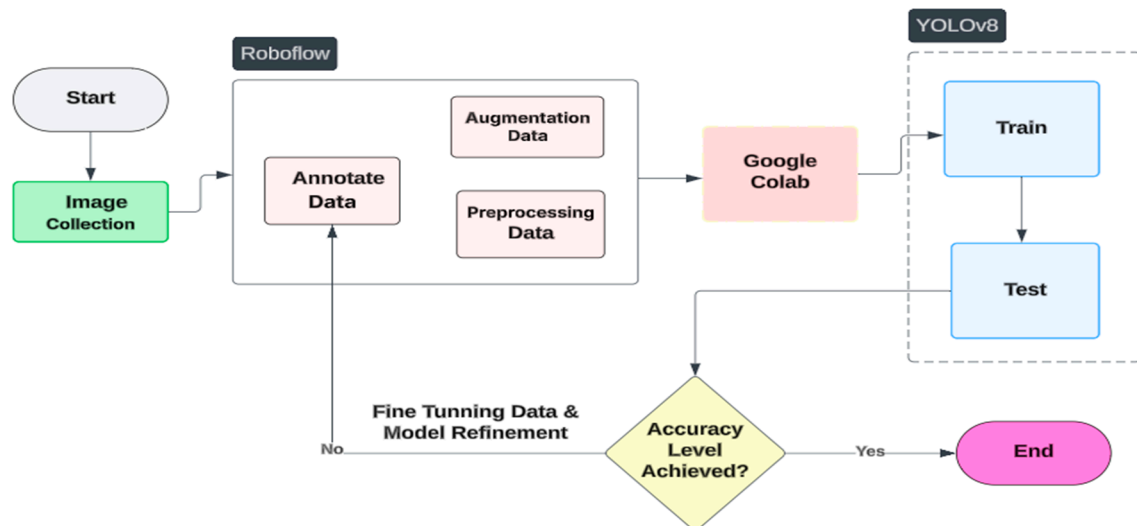


Figure 1. Flowchart of the methodology.

3.1. Image Collection

Several publicly available datasets, such as TrashNet [43] and TACO [44], are commonly used in deep learning for waste detection or classification research. However, this study developed a custom waste image dataset with 17 classes across three categories—non-recyclable, recyclable, and hazardous, as summarized in Table 1. This was necessary due to the limitations of existing datasets; TrashNet includes only 6 categories, and TACO relies on crowdsourced annotations that can be unreliable. To address these issues and expand the range of waste types, images were collected manually and from various public online sources. The resulting dataset, containing 3775 images, was resized and preprocessed using the Roboflow tool. It was then randomly partitioned into training (70%), testing (20%), and validation (10%) sets, ensuring that the category distribution in each subset closely mirrors that of the overall dataset. Table 2 provides the total number of samples for each class, highlighting the distribution across the recyclable, non-recyclable, and hazardous categories. Figure 2 visualizes the allocation of these data classes across the training, testing, and validation sets, ensuring balanced representation within each subset.

Table 1. Waste classification overview.

Category	Examples	Characteristics
Recyclable Waste	Can, Glass Bottle, Paper, Plastic Bottle, Cardboard	Recyclable waste materials can be reprocessed into new products, helping to reduce resource consumption.
Non-Recyclable Waste	Scrap Plastic, Plastic Plate, Plastic Cup, Light Bulb, Food Bag, Plastic Straw, Plastic Lid	Non-recyclable waste cannot be effectively reprocessed, typically requiring disposal through landfills.
Hazardous Waste	Chemical Plastic Bottle, Chemical Can, Battery, Paint Bucket, Medicine	Hazardous waste contains harmful substances that require special handling and disposal to avoid environmental damage.

Table 2. Dataset composition by class.

Class	Total Samples	Category
Can	271	Recyclable
Glass Bottle	315	Recyclable
Scrap Plastic	189	Non-Recyclable
Paper	192	Recyclable
Plastic Plate	168	Non-Recyclable
Plastic Cup	180	Non-Recyclable

Table 2. Cont.

Class	Total Samples	Category
Chemical Plastic Bottle	191	Hazardous
Chemical Can	188	Hazardous
Plastic Bottle	200	Recyclable
Light Bulb	168	Non-Recyclable
Food Bag	248	Non-Recyclable
Battery	273	Hazardous
Plastic Straw	268	Non-Recyclable
Plastic Lid	198	Non-Recyclable
Paint Bucket	257	Hazardous
Cardboard	199	Recyclable
Medicine	270	Hazardous

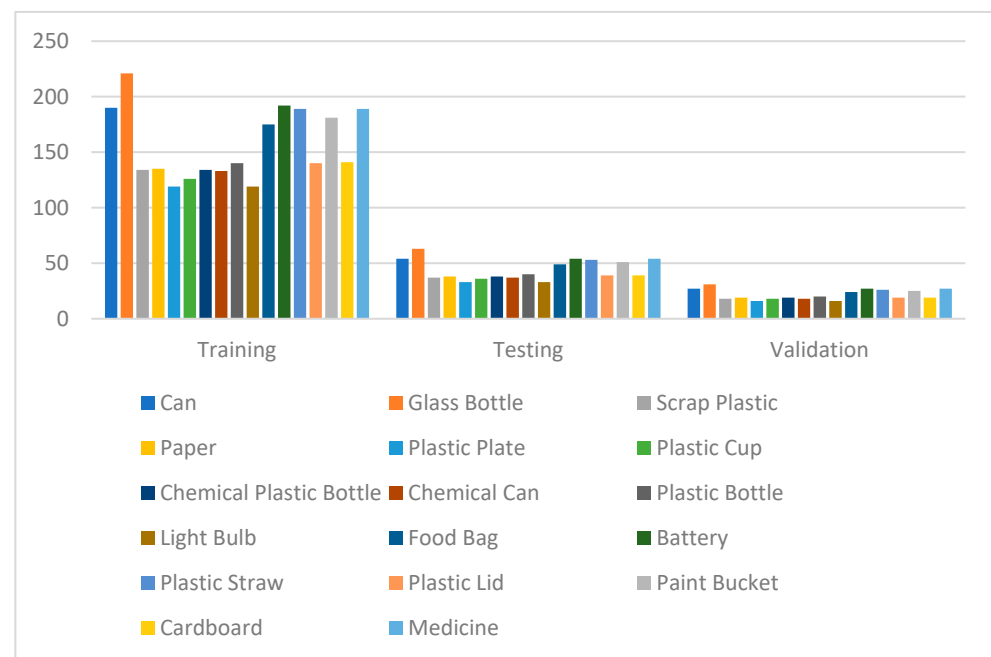


Figure 2. Distribution of data classes in training, testing, and validation.

3.2. Roboflow

Roboflow is a cloud-based platform designed to simplify the management, preparation, and deployment of datasets for computer vision projects. It plays a key role in waste detection by offering a one-stop solution for organizing, augmenting, and preprocessing image datasets. With its range of tools, Roboflow ensures that data preparation is efficient, making it easier to train high-performing machine learning models. In waste detection, well-prepared data are crucial for building effective models, and Roboflow helps streamline this process. The three main steps involved in preparing the dataset are the following:

1. **Data Annotation:** After collecting images, Roboflow is used to label objects in the images by defining bounding boxes or marking key features for the model to learn from. This crucial step ensures that the model can accurately identify and learn from the relevant features in each image.
2. **Data Preprocessing:** Roboflow applies preprocessing techniques like resizing, normalization, and noise reduction to ensure the images are optimized for training. This step is critical for improving model performance and accuracy by preparing the data in a form that enhances processing efficiency and effectiveness.
3. **Data Augmentation:** Before applying data augmentation, it is recommended to first train the model on the original waste dataset. This approach allows for an accurate

assessment of the model's baseline performance on real-world data and helps identify any underlying issues, such as class imbalance, inadequate representation, or lighting variations. If the model performs satisfactorily at this stage, significant augmentations might not be necessary, as they can introduce distortions or cause overfitting. Instead, if areas for improvement are identified, augmentations such as geometric transformations (rotating, flipping), color space adjustments (altering brightness, contrast), and introducing noise can be judiciously applied to specifically address these issues [45,46]. This strategy helps ensure that augmentations improve the model's robustness and ability to be generalized, rather than merely complicating the learning process.

3.3. Transfer to Google Colab

Once the dataset was prepared, it was transferred to Google Colab for model development and training. Google Colab provides a cloud-based platform with access to powerful hardware like GPUs and TPUs, making it ideal for training real-time waste detection models. It comes with essential Python libraries and machine learning frameworks, such as TensorFlow (<https://www.tensorflow.org/>, accessed on 8 November 2024), PyTorch (<https://pytorch.org/>, accessed on 8 November 2024), and OpenCV (<https://opencv.org/>, accessed on 8 November 2024), simplifying the process of setting up and training YOLOv8 models. For this research, Google Colab integrates smoothly with Roboflow, allowing annotated waste datasets to be directly imported using download codes, streamlining the process.

3.4. YOLOv8

YOLO is a cutting-edge approach in computer vision for object detection. Unlike traditional methods, where object detection and classification are typically performed as separate tasks, YOLO integrates both processes into a single, unified framework [23].

Object detection methods are crucial for various image-related tasks, and recent advancements have improved the ability to precisely localize regions of interest [47,48]. In the context of waste detection, where the accurate and efficient identification of different waste types is vital, current methods face challenges in balancing speed with accuracy. The YOLO variants, offering real-time detection capabilities, have proven effective in managing these demands [24]. The development of YOLO models from YOLOv1 to YOLOv8 has included ongoing improvements aimed at increasing detection accuracy and simplifying the models [49].

YOLOv8, released in 2023, is one of the latest versions of the YOLO object detection model, as shown in Figure 3. While it maintains the foundational architecture principles of its earlier versions, YOLOv8 introduces several key improvements. Unlike two-stage models, YOLOv8 simplifies the object detection process by directly predicting bounding boxes and class probabilities without requiring a separate region proposal network. One notable improvement in YOLOv8 is the adoption of an anchor-free, center-based approach, offering distinct advantages over the traditional anchor-based methods used in YOLOv5, YOLOv6, and YOLOv7. Additionally, YOLOv8 introduces Pseudo Ensemble or Pseudo Supervision (PS), which involves training multiple models with different configurations on the same dataset, enhancing prediction diversity and boosting accuracy and robustness. The model utilizes the Darknet-53 architecture, a 53-layer deep convolutional neural network optimized for feature extraction and object detection. A significant architectural upgrade is the replacement of the C3 module with the more efficient C2F module, improving both accuracy and processing times. YOLOv8 also replaces the 6×6 convolutional layer with a 3×3 layer in the model backbone, reducing the number of parameters and creating a more compact, computationally efficient network. Furthermore, it employs a decoupled head,

separating object detection from classification tasks, enhancing both speed and accuracy. These refinements make YOLOv8 a powerful solution for object detection and instance segmentation in computer vision applications.

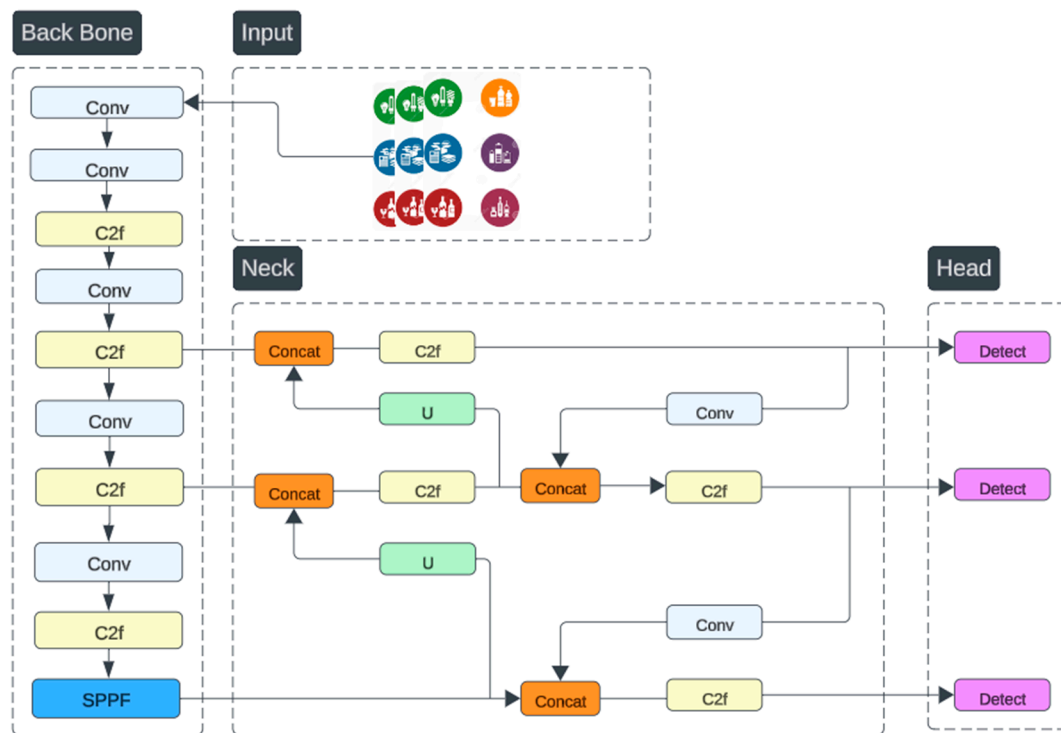


Figure 3. YOLOv8 architecture.

Inside Google Colab, the model training begins by using the YOLOv8 architecture. After training, the model is tested to evaluate its performance. This involves running the model on a test dataset to see how well it performs on unseen data. The model's performance is checked based on the desired accuracy level. Once the desired accuracy level is achieved, the process is completed, and the project is considered successful.

4. YOLOv8 Enhancement with Attention Mechanisms

This research advocates for incorporating attention mechanisms into YOLOv8's structure to mitigate the impact of background noise and boost the model's capability to identify and process features of small wastes, thereby increasing the accuracy of small waste target detection.

Attention mechanisms have shown substantial enhancements in performance across various visual intelligence tasks by efficiently refining relevant features [50,51]. In the field of object detection, squeeze and excitation (SE) [52] and the Convolutional Block Attention Module (CBAM) [53,54] are two prevalent attention mechanisms. SE focuses on modeling inter-channel dependencies to enhance the model's assessment of the importance of various channel features, although it overlooks their spatial distribution. On the other hand, CBAM expands on SE by integrating a spatial attention mechanism that improves the model's handling of positional details and local features.

The placement of the SE block and CBAM in the modified YOLOv8 models was strategically chosen based on insights from the original authors. While the optimal placement can vary and typically requires empirical validation, it is generally understood that placing attention mechanisms immediately after a concat operation may lead to suboptimal performance [52]. Accordingly, in my approach, both the SE block and CBAM are positioned after each C2f layer in the network's neck in the separate YOLOv8-SE and YOLOv8-CBAM

configurations. This placement is intended to enhance the attention focused on features extracted at these stages. Figures 4 and 5 illustrate the overall architecture of the YOLOv8 models with SE blocks and the CBAM, respectively, providing a clear visual reference of these modifications.

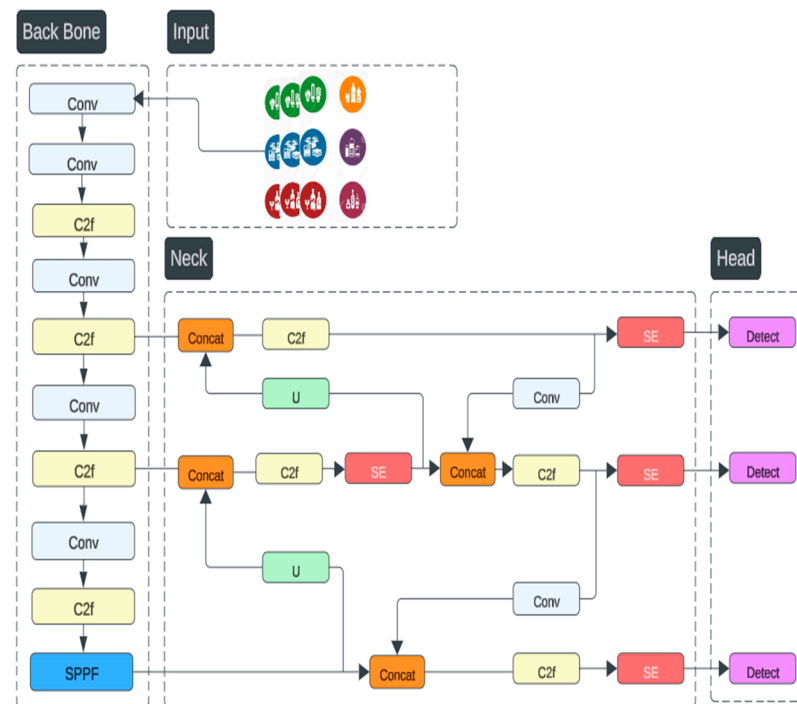


Figure 4. The proposed YOLOv8 with SE block modules.

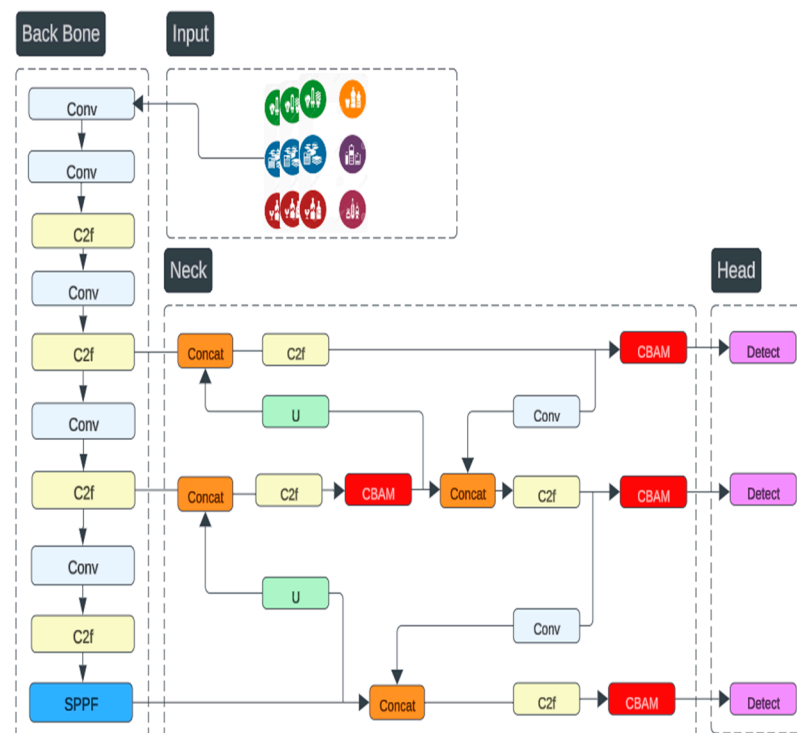


Figure 5. The proposed YOLOv8 with CBAMs.

4.1. Integration of SE Block Modules

The squeeze-and-excitation (SE) attention module is designed to boost the effectiveness of convolutional neural networks (CNNs) by emphasizing critical feature maps with only

a minimal increase in parameters. The SE attention mechanism involves two primary processes: squeeze and excitation. In the squeeze phase, global average pooling condenses the information across all feature maps into a channel-wise summary [52]. During the excitation phase, this summary is processed through two fully connected layers featuring non-linear activations and a sigmoid activation at the end [52]. The computed scale values from this process are then applied back to the original feature maps to enhance the CNN's focus on pertinent features [52]. Further details on the SE attention architecture are depicted in Figure 6.

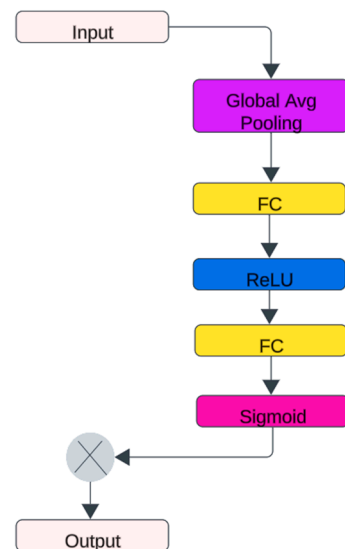


Figure 6. SE attention.

4.2. Integration of CBAM Modules

The Convolutional Block Attention Module (CBAM) is an innovative attention mechanism within convolutional neural networks that integrates both channel and spatial attentions [55]. This module actively adapts to the channel and spatial data from input feature maps to significantly enhance network performance [53,56]. CBAM is particularly noted for its superior effectiveness compared to other attention mechanisms, as illustrated in Figure 7.

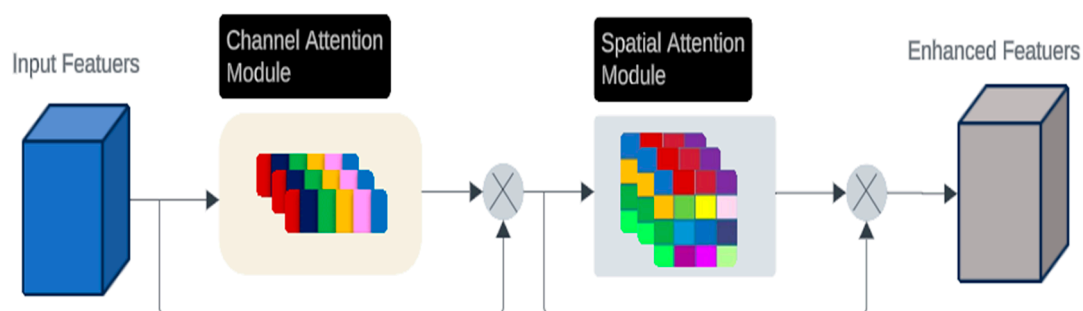


Figure 7. Schematic representation of the CBAM attention mechanism.

The operational process of the CBAM starts with the input feature map undergoing global average pooling across the channel dimension. It then progresses through two fully connected layers, where it analyzes the relationships and importance of each channel independently. Following this, the feature map is further processed with max-pooling along the channel dimension and then through two convolutional layers. This stage is crucial for determining the significance of various spatial positions within the feature map, which helps refine the focus on relevant areas.

Finally, the attention weights generated from the channel and spatial attention processes are separately applied to the input feature map. These weights are then fused via element-wise multiplication to produce the final feature representation. This advanced method effectively enhances the feature map by emphasizing the most relevant features, thus improving the accuracy and efficiency of the neural network's performance without a substantial increase in computational complexity.

5. Experiments

5.1. Hardware and Equipment

This research will utilize a computer and a camera to create the proposed real-time deep learning detection and classification model. The computer, featuring an AMD Ryzen 7 processor, an NVIDIA GeForce RTX 3050 graphics card, and 16 GB of RAM, will act as the central hub of the system. It will handle code execution and support model training and testing, ensuring the model remains robust and reliable. At the same time, the camera will be essential for validating the object detection model, allowing for real-time monitoring by smoothly integrating with the Windows 11 operating system.

5.2. Performance Metrics

This paper uses several key metrics to assess the model's performance in detecting and classifying waste items, including accuracy, precision, recall, F1-score, mean average precision (mAP), bounding box loss, classification loss, and distribution focal loss (DFL). Precision (P) and recall (R) are outlined in Equations (1) and (2), while the F1-score, balancing precision and recall, is shown in Equation (3):

$$P = \frac{TP}{TP + FP} \quad (1)$$

$$R = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = \frac{2PR}{P + R} \quad (3)$$

where TP represents the number of correctly predicted samples (true positives), FP refers to incorrectly predicted positive samples (false positives), and FN denotes undetected targets (false negatives). The average precision (AP) defines the average precision value for recall values over 0 to 1. The mean average precision (mAP) is the average of AP values across all classes, as shown in Equations (4) and (5):

$$AP = \int_0^1 P(x) dx \quad (4)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP_i \quad (5)$$

When handling classification challenges, especially those aimed at accurately classifying a particular category, the task is often approached as a binary classification problem. This involves dividing the classification targets into two groups: positive and negative. In these scenarios, a confusion matrix is frequently utilized to depict the performance of the model. The confusion matrix is a table that displays how the model's predictions align with the actual labels, providing a straightforward assessment of the model's effectiveness in identifying both positive and negative cases. The use of a confusion matrix is fundamental in binary classification tasks, as depicted in Table 3.

Table 3. Confusion matrix.

Confusion Matrix		True Value	
		Positive	Negative
Predicted Value	Positive	TP	FP
	Negative	FN	TN

In addition to these classification metrics, the model's performance is also assessed using bounding box loss, classification loss, and distribution focal loss, which evaluate object localization and classification accuracy. Bounding box loss measures the difference between predicted and ground truth bounding box coordinates, often using Smooth L1 loss, as shown in Equation (6):

$$L_{box} = Smooth_{L1}(\hat{b} - b) \quad (6)$$

where $\hat{b}$ represents the predicted bounding box coordinates and b represents the ground truth coordinates. The classification loss is calculated using cross-entropy loss, which measures how well the model predicts the correct class labels, as shown in Equation (7):

$$L_{cls} = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (7)$$

Finally, distribution focal loss refines localization precision by assigning greater weight to well-classified samples, improving detection accuracy for smaller or harder-to-detect objects. The DFL is defined in Equation (8):

$$L_{dfl} = -\alpha(1 - \hat{p})^\gamma \log(\hat{p}) \quad (8)$$

where α is a balancing factor, $\hat{p}$ is the predicted probability for the true class, and γ is a focusing parameter to reduce the impact of easy-to-classify samples. The total model loss is a weighted sum of these individual components, ensuring the model performs well in both object localization and classification tasks.

These metrics collectively ensure a comprehensive evaluation of the model's performance, encompassing its ability to accurately classify waste items, precisely localize them within images, and maintain overall effectiveness across different waste categories.

5.3. Training Analysis of YOLOv8

The training process extended across 200 epochs, with the model's performance evaluated on both the training and validation sets using key metrics, including bounding box loss (box_loss), classification loss (cls_loss), distribution focal loss (dfl_loss), precision, recall, and mean average precision (mAP) across multiple intersection over union (IoU) thresholds. Figure 8 presents the results in graph form. An epoch represents a full pass through the entire image dataset in YOLOv8. During each epoch, the YOLOv8 model processes all bounding boxes in the dataset and adjusts its parameters using the specified loss function and optimization algorithm.

The decreasing trends in both training and validation losses, coupled with the steady improvement in precision, recall, and mAP metrics, indicate that the model effectively learned to detect and classify waste objects across the three waste categories. The near alignment of the training and validation loss curves suggests minimal overfitting, while the consistently high precision, recall, and mAP scores confirm the model's accuracy in both object localization and classification. These results suggest that YOLOv8 is highly suitable for automated waste detection, offering high detection accuracy and generalization capability across different waste types.

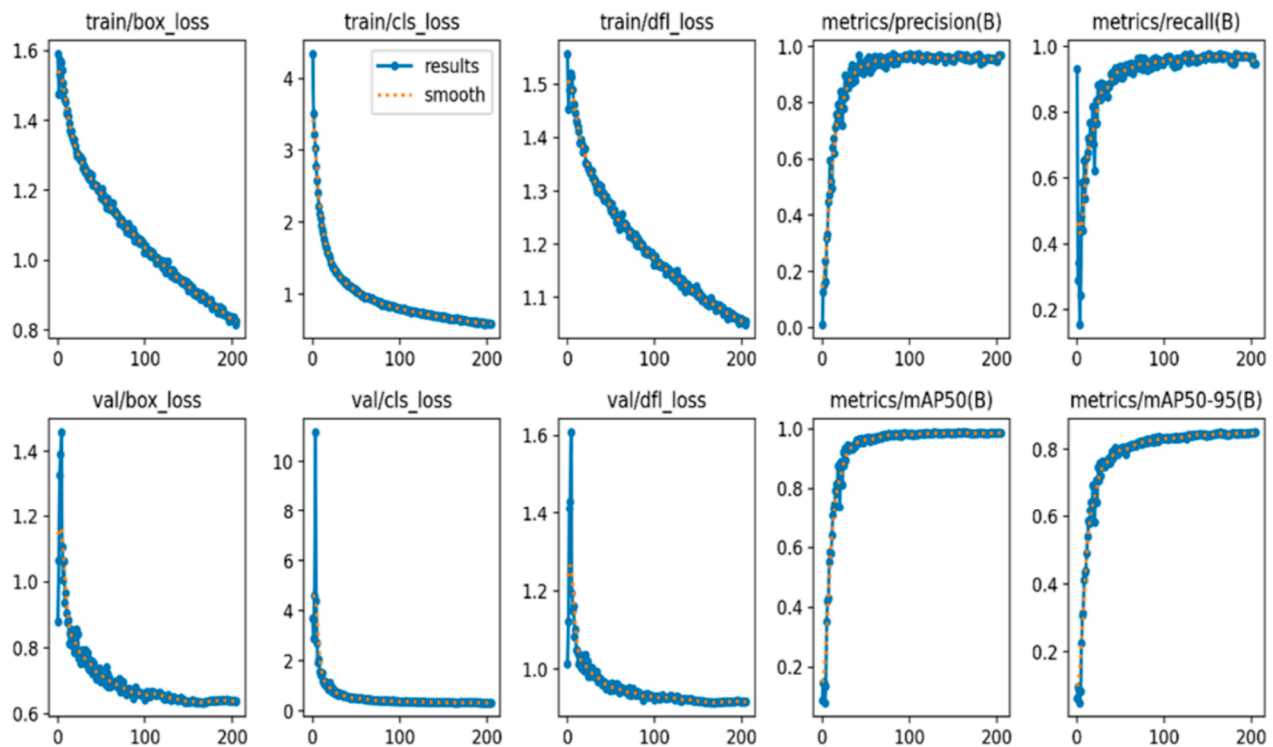


Figure 8. The results of the model training and validation.

5.4. Evaluation Results

After completing the training and validation phases, the YOLOv8 model was tested on a separate, unseen test set. A sample of the waste identification results obtained by using the well-trained YOLOv8 model is shown in Figure 9.

Several performance metrics were employed to evaluate the model, including mAP at a 50% IoU threshold, precision, recall, and the confusion matrix. The confusion matrix provides further insight into the model's performance by highlighting the specific instances where misclassifications occurred, allowing for a detailed analysis of how the model distinguishes between different waste types. The evaluation results, including precision, recall, and mAP, are shown in Table 4, while the confusion matrix is presented in Figure 10.

Table 4. Model performance metrics.

Class	Precision	Recall	mAP
Can	0.89	0.89	0.84
Glass Bottle	0.98	0.91	0.95
Scrap Plastic	0.86	0.81	0.74
Paper	0.91	0.82	0.79
Plastic Plate	0.85	0.85	0.77
Plastic Cup	0.94	0.82	0.82
Chemical Plastic Bottle	0.95	0.86	0.87
Chemical Can	0.94	0.91	0.91
Plastic Bottle	0.89	0.93	0.88
Light Bulb	0.83	0.93	0.82
Food Bag	0.80	0.95	0.81
Battery	0.89	0.96	0.91
Plastic Straw	0.94	0.88	0.88
Plastic Lid	0.89	0.86	0.81
Paint Bucket	0.87	0.9	0.83
Cardboard	0.90	0.97	0.93
Medicine	0.93	0.96	0.95



Figure 9. Sample of the model recognition results.

The YOLOv8-based waste detection and classification model shows strong performance across several key classes, particularly in identifying recyclable and hazardous waste types. For recyclable waste, the model demonstrates high precision, recall, and mAP values, reflecting its ability to accurately classify these materials. For instance, Glass Bottle achieves a precision of 0.98, a recall of 0.91, and an mAP of 0.95, underscoring the model's accuracy in identifying glass waste. Can also exhibits balanced precision (0.89) and recall (0.89), with an mAP of 0.84, making it one of the more reliably detected waste

types. Cardboard stands out with a recall of 0.97 and a precision of 0.90, showing that the model excels in distinguishing this recyclable material from others. These high values in recyclable categories reflect the model's efficacy in real-time waste sorting systems, where accurate classification is critical.

Confusion Matrix

	Can	Glass Bottle	Scrap Plastic	Paper	Plastic Plate	Plastic Cup	Chemical Plastic Bottle	Chemical Can	Plastic Bottle	Light Bulb	Food Bag	Battery	Plastic Straw	Plastic Lid	Paint Bucket	Cardboard	Medicine
Can	48	0	0	0	1	0	1	0	0	1	1	0	0	0	0	1	1
Glass Bottle	1	49	0	0	1	0	1	0	0	0	0	0	0	0	2	0	0
Scrap Plastic	1	0	30	0	0	1	0	0	1	1	2	0	0	1	0	0	0
Paper	0	0	0	32	0	0	0	0	1	0	0	0	1	2	2	1	0
Plastic Plate	0	0	0	0	29	0	0	0	0	1	1	1	1	0	0	0	1
Plastic Cup	1	0	1	2	1	33	0	0	0	0	0	0	0	0	0	1	1
Chemical Plastic Bottle	0	0	2	0	0	0	36	0	0	1	3	0	0	0	0	0	0
Chemical Can	1	0	0	0	0	0	0	30	0	1	0	0	1	0	0	0	0
Plastic Bottle	0	0	1	0	0	0	0	0	40	0	2	0	0	0	0	0	0
Light Bulb	0	0	0	0	0	0	0	0	0	25	0	1	0	0	1	0	0
Food Bag	0	0	0	0	0	0	0	1	0	0	41	1	0	0	0	0	0
Battery	1	0	0	0	0	0	0	0	0	0	0	51	0	0	0	0	1
Plastic Straw	0	1	0	1	1	0	0	0	1	0	1	0	46	0	0	1	0
Plastic Lid	0	0	0	0	1	0	0	0	2	0	0	1	0	32	1	0	0
Paint Bucket	1	0	1	0	0	0	0	1	0	0	2	0	0	0	46	0	0
Cardboard	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	36	0
Medicine	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	51

Predicted Class

Figure 10. Evaluation of the model performance in the form of a confusion matrix.

However, insights from the confusion matrix reveal challenges with misclassifications in certain categories, especially those that share visual similarities. For example, while Scrap Plastic achieves a precision of 0.86 and recall of 0.81, its mAP drops to 0.74 due to frequent misclassification into other plastic-related categories like Plastic Cup and Plastic Lid. This issue is highlighted in the confusion matrix, where overlaps between these classes are evident. Such confusion arises from the visual similarity between plastic items, leading the model to struggle in differentiating between Scrap Plastic and other non-recyclable plastic waste types. A similar trend is seen with Plastic Plate and Plastic Cup, where the confusion matrix shows misclassifications between these two categories. Despite strong precision for Plastic Cup (0.94), the recall of 0.82 indicates that the model occasionally confuses it with Plastic Plate or Scrap Plastic, negatively impacting its overall performance.

For hazardous waste, the model performs exceptionally well. Battery achieves high recall (0.96) and precision (0.89), making it a well-detected category, as confirmed by the confusion matrix, which shows minimal misclassifications. Chemical Can and Chemical Plastic Bottle also show strong performance, with precision values of 0.94 and 0.95, respectively, and very few occurrences of confusion with other waste types. The confusion matrix reveals that hazardous waste categories have distinct features that the model identifies effectively, resulting in fewer misclassifications compared to non-recyclable plastic waste. This high performance is essential in ensuring the safe handling of hazardous waste in real-time sorting systems, as precise detection is critical for avoiding health and environmental risks.

Despite the model's strong overall performance, certain non-recyclable and hazardous waste categories face some misclassification issues. Light Bulb achieves a recall of 0.93 but shows lower precision (0.83), as reflected in the confusion matrix, where it is often confused with Plastic Lid. Similarly, Food Bag, although having a high recall of 0.95, is frequently misclassified as Scrap Plastic or Plastic Plate, leading to a lower precision (0.80) and mAP (0.81). These issues, visible in the confusion matrix, highlight the challenge of

distinguishing between visually similar waste items, especially those with overlapping material properties.

The confusion matrix also provides insights into the model's struggle with underrepresented classes, such as Scrap Plastic and Plastic Lid, where limited data lead to higher misclassification rates. This suggests that increasing the dataset size for these categories or implementing more advanced techniques, such as data augmentation or synthetic data generation, could improve model performance.

To focus on the model's performance at a higher categorical level, the confusion matrix for the YOLOv8-based waste classification model is aggregated into three main categories—recyclable, non-recyclable, and hazardous. Figure 11 shows the confusion matrix for the higher categorical level; this aggregation simplifies the analysis by highlighting broader classification trends, making it easier to assess the model's overall accuracy in distinguishing between major waste categories. It is particularly useful for understanding how well the model handles broad sorting applications in real-world scenarios, such as waste management systems that primarily need to separate materials based on whether they are recyclable or hazardous. By grouping similar classes, such as various types of plastics or hazardous items, the analysis can better reveal overarching patterns and potential issues in classification that might be masked at the individual class level.

Confusion Matrix for Recyclable, Non-Recyclable, and Hazardous Categories

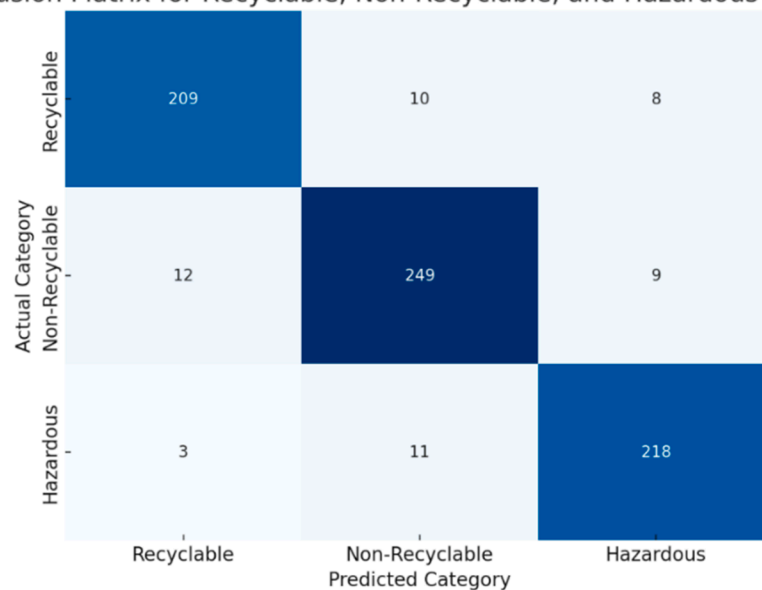


Figure 11. The confusion matrix for the higher categorical level.

When evaluating the aggregated confusion matrix, the model demonstrates strong performance across all three categories, with precision, recall, and F1-scores exceeding 90%, reflecting its accuracy in both detecting and classifying waste. The chart in Figure 12 compares these evaluation metrics for the three aggregated waste categories. This visualization provides a concise overview of the model's effectiveness in detecting and classifying waste into these three categories, highlighting its consistent accuracy across all metrics.

Moreover, the results of these metrics, combined with the model's overall accuracy of 92.73%, demonstrate the model's effectiveness in handling broad waste categories. The formula for calculating accuracy is as follows:

$$Accuracy = \frac{\sum TP}{Total\ Test\ Samples} \quad (9)$$

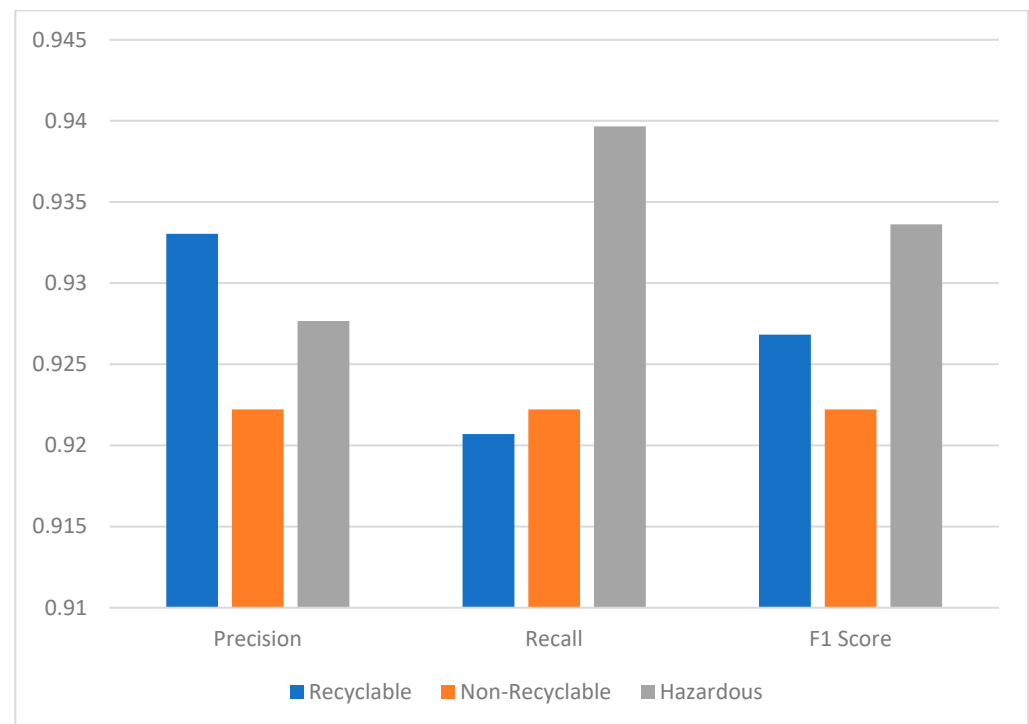


Figure 12. Precision, recall, and F1-score comparison for the three waste categories.

In the aggregated confusion matrix, the classes are grouped into broader categories (recyclable, non-recyclable, hazardous). This means that misclassifications within each category (e.g., “Can” misclassified as “Glass Bottle”) are no longer counted as errors in the aggregated matrix, as they still fall under the same category (recyclable waste).

When comparing the model’s accuracy at the aggregated level (92.73%) with the accuracy of the class-level confusion matrix (89.85%), the higher accuracy at the aggregated level indicates that the model is more successful at distinguishing between broad categories than between specific individual waste types.

Although the aggregation improves overall accuracy by simplifying the classification into major categories, it sacrifices some detail regarding class-level performance. This approach allows us to focus on reducing major category-level misclassifications, such as confusing recyclable materials with hazardous waste, which is crucial for real-world waste processing. However, future refinements could further improve precision and recall by addressing specific misclassifications within these broader categories and individual waste types.

5.5. Comparative Analysis of the Original YOLOv8 and Other Deep Learning Model

To further assess the performance of the proposed YOLOv8-based detection and classification model for household waste, a comparison was conducted against other state-of-the-art deep learning models. Table 5 presents the results, demonstrating that YOLOv8 achieves the highest mean mAP at 85.3%, surpassing all other models.

Among the models compared, YOLOv7 follows closely with an mAP of 82.1%, showing that while both YOLO versions are highly effective, the latest iteration provides a noticeable improvement in accuracy. The EfficientDet model achieves 80.4% mAP, reflecting competitive performance, but still lags behind YOLOv8 by 4.9%, indicating that YOLOv8’s optimization techniques enhance detection accuracy. RetinaNet, scoring 81.6%, also performs well but remains 3.7% lower than YOLOv8, suggesting that the one-stage detection approach in YOLOv8 provides an advantage in precision.

Table 5. Comparative performance of detection models based on mAP%.

Model	mAP %
YOLOv8	85.3
YOLOv7	82.1
SDD [57]	78.2
EfficientDet [58]	80.4
RetinaNet [59]	81.6
Fast R-CNN [60]	77.3
Mask R-CNN [61]	74.1

Models such as SDD (78.2%), Fast R-CNN (77.3%), and Mask R-CNN (74.1%) show significantly lower mAP scores, with YOLOv8 outperforming them by 7.1%, 8%, and 11.2%, respectively. This performance gap highlights the efficiency of YOLOv8 in detecting and classifying household waste compared to older region-based CNN models, which often struggle with real-time processing due to their multi-stage architecture.

Overall, these results reinforce that YOLOv8 provides superior accuracy and efficiency in waste classification compared to previous YOLO versions and other *deep learning* models. The combination of its lightweight architecture, real-time detection capabilities, and high precision makes YOLOv8 the most effective choice for household waste classification tasks.

5.6. Evaluation of the Impact of Data Augmentation and Attention Mechanisms

To investigate the effects of data augmentation and attention mechanisms on waste classification and detection performance, five augmentation techniques were applied: rotating, flipping, blurring, random cropping, and darkening. These techniques are expected to enhance dataset quality by addressing variations in orientation, lighting, visibility, and environmental conditions, thereby improving model generalization. As a result, the dataset size increases from 3775 to 5695 samples, ensuring a more balanced distribution for training. The number of image samples before and after augmentation is analyzed and presented in Table 6.

Table 6. Number of samples per class before and after data augmentation.

Class	Original Samples	Augmented Samples	Total Samples After Augmentation
Can	271	64	335
Glass Bottle	315	20	335
Scrap Plastic	189	146	335
Paper	192	143	335
Plastic Plate	168	167	335
Plastic Cup	180	155	335
Chemical Plastic Bottle	191	144	335
Chemical Can	188	147	335
Plastic Bottle	200	135	335
Light Bulb	168	167	335
Food Bag	248	87	335
Battery	273	62	335
Plastic Straw	268	67	335
Plastic Lid	198	137	335
Paint Bucket	257	78	335
Cardboard	199	136	335
Medicine	270	65	335
Total	3775	1920	5695

Furthermore, the introduction of SE and CBAM attention mechanisms was tested to thoroughly validate the effectiveness of the proposed improvements. To separately

evaluate the contributions of the three modules, data augmentation, YOLOv8-SE, and YOLOv8-CBAM were tested independently. Then, to further examine their combined effects, experiments were conducted using data augmentation with the original YOLOv8 model, data augmentation with YOLOv8-SE, and data augmentation with YOLOv8-CBAM. A total of six different experiments were carried out to systematically assess the impact of each configuration on the model's performance. The results of these experiments are detailed in Table 7.

Table 7. Impact of data augmentation and attention mechanisms on YOLOv8 model performance.

Models	Data Augmentation	SE Block	CBAM	mAP%
YOLOv8	×	×	×	85.3
	✓	×	×	87.4
	×	✓	×	87.1
	×	×	✓	88.2
	✓	✓	×	88.7
	✓	×	✓	89.5

Note: a checkmark ('✓') indicates that this improvement was implemented, and a cross ('×') indicates that this improvement was not implemented.

The evaluation of data augmentation and attention mechanisms using YOLOv8 demonstrates significant improvements in waste classification accuracy. The baseline YOLOv8 model achieves an mAP of 85.3%, serving as a reference for performance assessment. When data augmentation is applied, the mAP increases to 87.4%, indicating a substantial improvement. This improvement is due to the enhanced generalization from applied augmentation techniques and due to balancing the dataset. Data augmentation helps to mitigate class imbalance by artificially increasing the representation of underrepresented classes, which enhances the model's ability to generalize across varied inputs.

Attention mechanisms further enhance performance. The integration of SE blocks (YOLOv8-SE) raises the mAP to 87.1%, suggesting that channel-wise recalibration effectively enhances feature extraction. However, the YOLOv8 model with the CBAM achieves an mAP of 88.2%, outperforming the SE block by 1.1%. This indicates that the combination of spatial and channel attention in CBAM helps the model better focus on relevant features while minimizing background distractions.

Combining data augmentation with SE blocks results in an mAP of 88.7%, reflecting a moderate improvement over using SE blocks alone. This combination indicates that data augmentation and channel attention are complementary. The best performance is achieved with YOLOv8 with both data augmentation and CBAM, reaching an mAP of 89.5%, which is a 4.2% increase over the baseline model. This confirms that the dual enhancement from CBAM and data augmentation, particularly in balancing the dataset, maximizes classification accuracy. Figure 13 displays the detection outcomes in a realistic waste scenario with background noise, showcasing the performance of YOLOv8, YOLOv8-SE, and YOLOv8-CBAM, all trained on the enhanced augmented dataset.

Overall, the CBAM consistently outperforms SE blocks, demonstrating the advantage of integrating both spatial and channel attention. While data augmentation alone provides significant improvements, its role in balancing the dataset through augmented samples and its combination with attention mechanisms, particularly CBAM, yields the most substantial gains in classification accuracy.

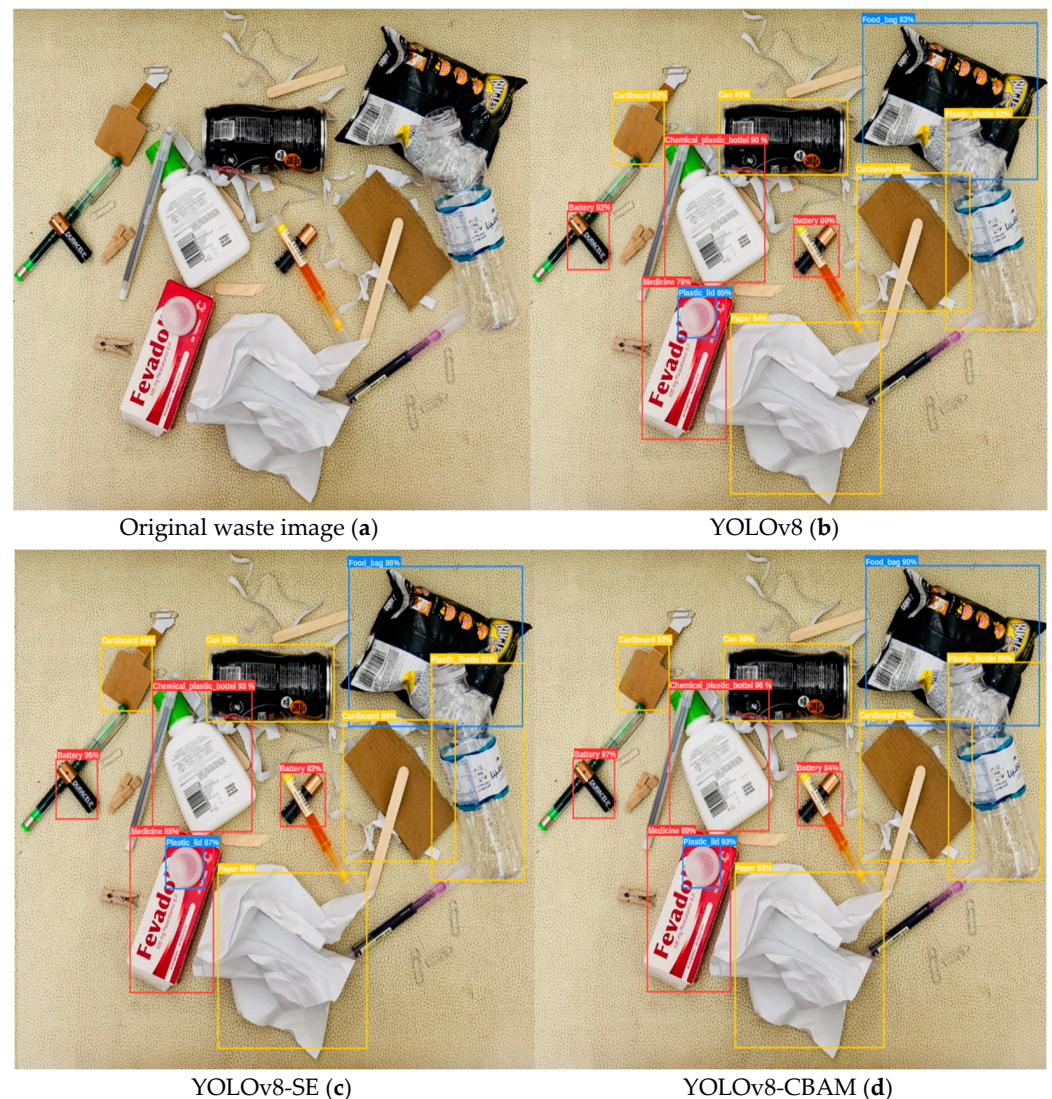


Figure 13. Comparison of performance. (a) original waste image; (b) result of detection by YOLOv8; (c) result of detection by YOLOv8-SE; (d) result of detection by YOLOv8-CBAM.

6. Conclusions

In conclusion, this research underscores the significant advancements in household waste management enabled by the proposed YOLOv8-based real-time waste detection and classification system. Leveraging the sophisticated capabilities of YOLOv8, alongside the integration of attention mechanisms and data augmentation strategies, this model notably enhances detection accuracy while operating in real-time. These enhancements mark a substantial improvement over traditional two-stage detectors. The YOLOv8's streamlined single-stage approach, augmented by the squeeze-and-excitation (SE) blocks and Convolutional Block Attention Module (CBAM), provides an effective, high-performance alternative that excels in real-world applications. While many studies focus on a limited number of waste classes, this study showcases the capabilities of YOLOv8 in providing an automated solution for detecting and classifying a diverse range of 17 waste classes, including hazardous, recyclable, and non-recyclable household items. Demonstrating a high mAP of 89.5%, the five data augmentation techniques and YOLOv8-CBAM model effectively manage the detection and classification of this extensive array of waste types.

The proposed framework demonstrates the significant potential of deep learning-based methods for household waste management by accurately categorizing various types of waste. This substantial expansion in the scope of automated waste detection and

classification solutions highlights the transformative potential of YOLOv8 in enhancing the efficiency and accuracy of waste management practices, thereby revolutionizing waste sorting systems. In the future, the integration of deep learning models into smart bins and edge devices could significantly enhance their functionality and efficiency in real-time waste management. This technology has the potential to enable the installation of robotic arms in waste management facilities, allowing them to autonomously identify and sort different classes of objects, thereby eliminating the need for human intervention. Such advancements could automate waste recycling processes both in industrial settings and households through the use of deep learning algorithms. This integration would make the recycling process more adaptive and accessible at a community level, promoting a more sustainable environment.

7. Future Work

As Artificial Intelligence endeavors to outperform human accuracy, enhancing the system's performance is a crucial future goal for the proposed system to work efficiently in industrial settings. It should be emphasized that industrial waste sorting often involves greater complexities compared to residential waste [15]. Consequently, for practical applications, it is essential to classify waste into more detailed categories, including distinct types of hazardous, recyclable, and non-recyclable waste. This includes various metal alloys, diverse chemical materials, different kinds of batteries, and multiple types of plastics. Employing synthetic data can present a significant advantage in expanding the training landscape for AI systems in complex settings [62–64]. This method can improve the accuracy and generalizability of detection models and expedite their development by minimizing the need for extensive real data collection.

To further advance these efforts, future work will also explore more sophisticated methodologies such as the Part-Object Relational Network [65], which utilizes capsule-based relational dynamics for enhanced recognition of mixed-material waste. Additionally, integrating Stacked Residual Self-Attention CNNs with Vision Transformers [66], techniques proven effective in remote sensing, could substantially improve waste segmentation under diverse environmental conditions. These innovative approaches promise significant enhancements in the accuracy and computational efficiency of smart recycling technologies, paving the way for more effective and efficient waste management solutions in industrial settings.

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